

Dat Nguyen

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EDUCATION

Michigan State University

B.S. in Computer Science | Honors College; Dean's List | GPA: 3.8

East Lansing, MI

Aug. 2022 – May 2026

EXPERIENCE

Software Engineer Intern

Hanoi, Vietnam (Remote)

DIIJAM - Electronic Music Streaming Platform

May 2025 – August 2025

- Built subscription interface in **Next.js/TypeScript** with comprehensive validation, enhancing user experience for **1M+ users and 15K+ DJs**.
- Created affiliate marketing system in **Python/Flask** with unique discount codes and commission tracking, enabling **500+ DJs** to generate recurring revenue through their audience
- Streamlined payment verification by integrating **Gmail API** to automatically parse bank transfer confirmations, eliminating **8+ hours/week** of manual finance team work with near perfect accuracy.

Software Engineering Intern

San Francisco, CA (Remote)

nuoc.solutions

May 2024 – May 2025

- Developed water quality monitoring platform using **Next.js/FastAPI/PostgreSQL** to track data from **200+ stations**, providing **1000+ farmers** with real-time insights through **Mapbox and Chart.js** visualizations.
- Integrated LLM-powered analysis allowing farmers to query water conditions in natural language, processing **500+ monthly queries** and delivering actionable recommendations.
- Containerized and deployed services with **Docker** on **Azure**, successfully scaling the monitoring platform and citizen-reporting portal across **multiple provinces**.

Software Engineering Intern

Hanoi, Vietnam

HCLTech

May 2023 – Aug. 2023

- Set up static application security testing (SAST) in CI/CD pipelines using **Python/Pylint** and **GitHub Actions**, automating security scans of **10K+ server files** and cutting manual review time by **70%**.
- Built anomaly detection pipeline using **Isolation Forest** and **One-Class SVM** algorithms, successfully flagging **150+ high-risk security events** with **95% precision**.
- Implemented real-time file integrity monitoring (FIM) with automated alerting, reducing mean time to detect security incidents by **40%** across **200+ systems**.

PROJECTS

Real-Time Event-Driven Crypto Trading Platform | *Java 21, Spring Boot, Redis, Docker*

- Built a low-latency, event-driven Java backend using **Spring WebFlux**, **Project Reactor**, and **Redis Streams** to ingest and process real-time WebSocket market data at scale.
- Developed an autonomous AI Trading Agent integrating OpenAI's LLM, capable of interpreting natural language commands
- Designed a reactive microservices architecture with clear service boundaries and transactional data pipelines in **PostgreSQL**, supporting concurrent workloads, system extensibility, and fault isolation.
- Delivered production-grade infrastructure including secure REST APIs (JWT), Redis-based messaging, and fully containerized services with **Docker**, enabling reproducible, cloud-ready deployment.

AI Travel Agent | *Next.js, Node.js, TypeScript, Tailwind*

- Created AI travel planning system that collects user preferences and generates personalized itineraries, cutting planning time by **80%** for **50+ test users** compared to manual research.
- Built **Node.js** backend integrating multiple travel APIs with LLM orchestration, implementing **CI/CD** pipeline with comprehensive test coverage at **95%**.
- Designed responsive **Next.js/TypeScript** frontend with intuitive chat interface, handling complex state management for **100+ concurrent user sessions**.

TECHNICAL SKILLS AND RELEVANT COURSEWORK

Programming Languages: Python, TypeScript/JavaScript, Java, C++, C, SQL, HTML/CSS

Frameworks & Tools: Node.js, Express, FastAPI, Flask, Django, React, Next.js, Tailwind CSS, PyTorch, scikit-learn, LLM APIs, REST APIs, Docker, GitHub Actions, Git, PostgreSQL, RabbitMQ, MySQL, Linux, AWS, Azure

Relevant Coursework: Data Structures & Algorithms, Cloud Computing, Big Data Analysis, Computer Architecture, Object-Oriented Programming